Business Question and Visualization Report

Date	01 November 2026
Team ID	
Project Name	Health360: Comprehensive Healthcare Risk & Disease Analysis
Maximum Marks	5 Marks

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting

Business Questions and Visualisation

The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations in Tableau. Well-framed questions help in identifying key metrics, selecting relevant data, and building visualisation that provide insights.

1. How many patients are in the dataset, and how many have diabetes, heart disease or a stroke history?

Visualization: KPI cards for total patients, diabetes, heart disease and stroke.

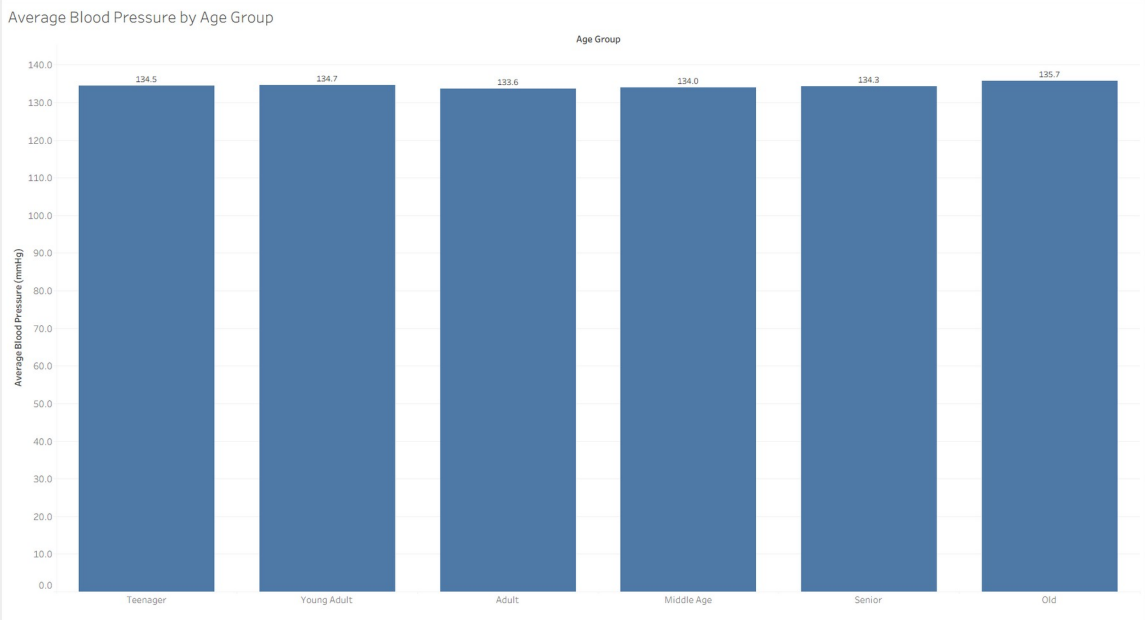
Insight: There are 5,000 patients: 1,746 (34.9%) have diabetes, 1,499 (30.0%) have heart disease and 1,243 (24.9%) have had a stroke.

Total Patients	Diabetes Patients	Heart Disease Patients	Stroke Patients
5,000	1,746	1,499	1,243

2. How does average blood pressure vary across age groups?

Visualization: Bar chart of average blood pressure by age group.

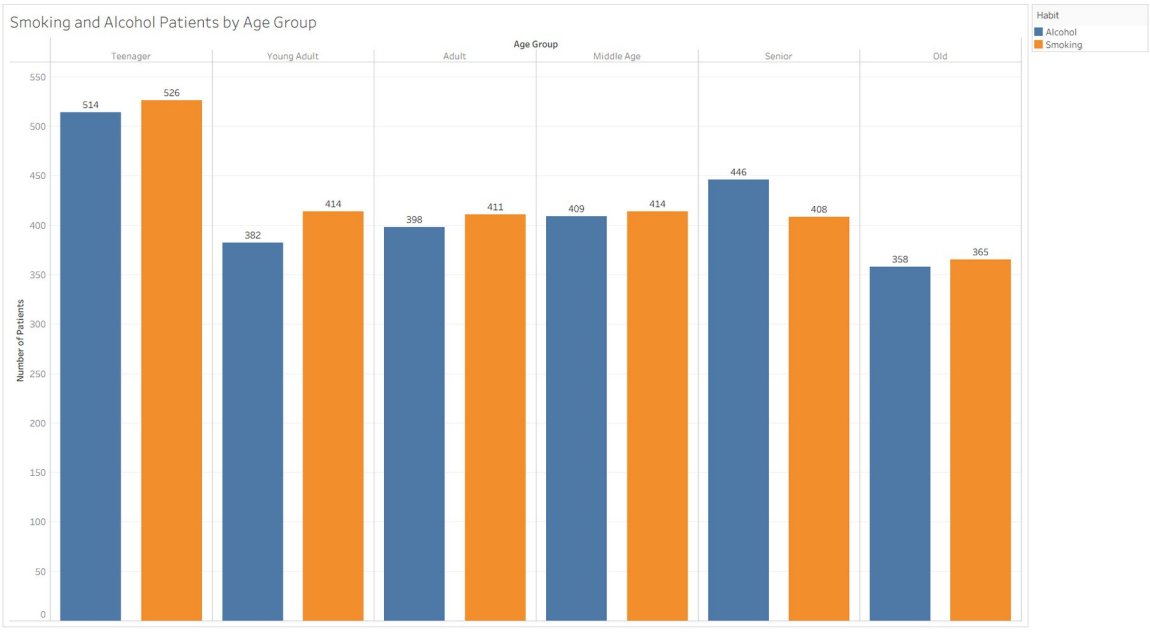
Insight: Averages sit between 133.6 mmHg (Adult) and 135.7 mmHg (Old), a gap of only about 2 mmHg, so high blood pressure is common in every age group.



3. How do smoking and alcohol habits differ across age groups?

Visualization: Grouped bar chart of smoking and alcohol patients by age group.

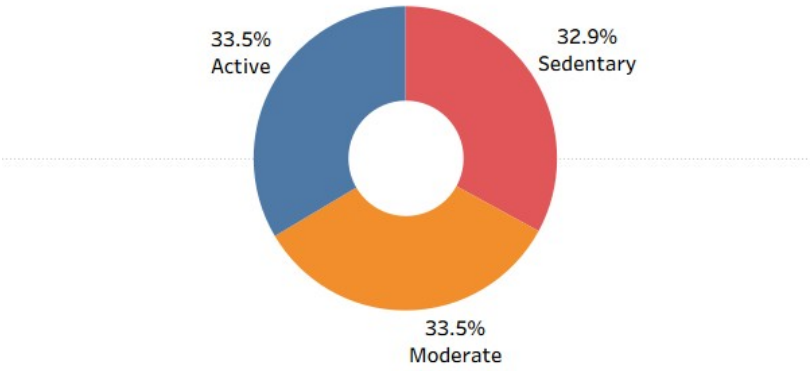
Insight: The Teenager group (18 to 30) has the most smokers (526) and alcohol users (514); the Old group has the fewest (365 and 358).



4. How are patients distributed across exercise levels?

Visualization: Donut chart of patients by exercise level.

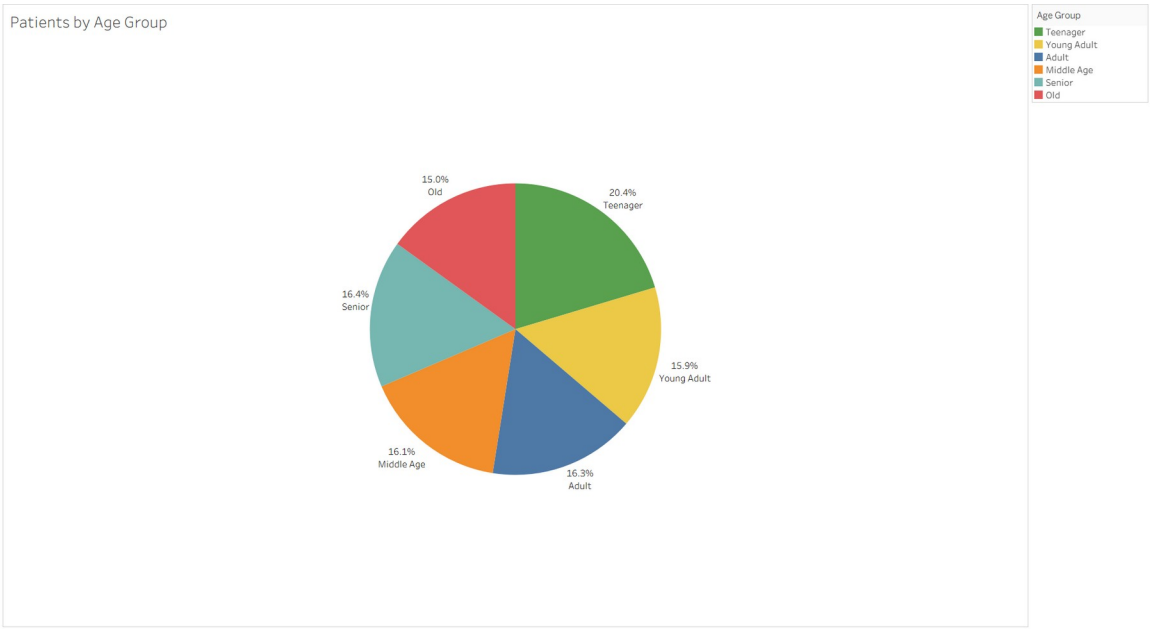
Insight: Moderate and Active are each 33.5% and Sedentary is 32.9%, so the population is split almost evenly.



5. What is the age mix of the patient base?

Visualization: Pie chart of patients by age group.

Insight: Teenagers are the largest group at 20.4%; the other five groups range from 15.0% to 16.4%.



6. How do heart disease and diabetes counts vary across BMI categories?

Visualization: Grouped bar chart of heart disease and diabetes patients by BMI category.

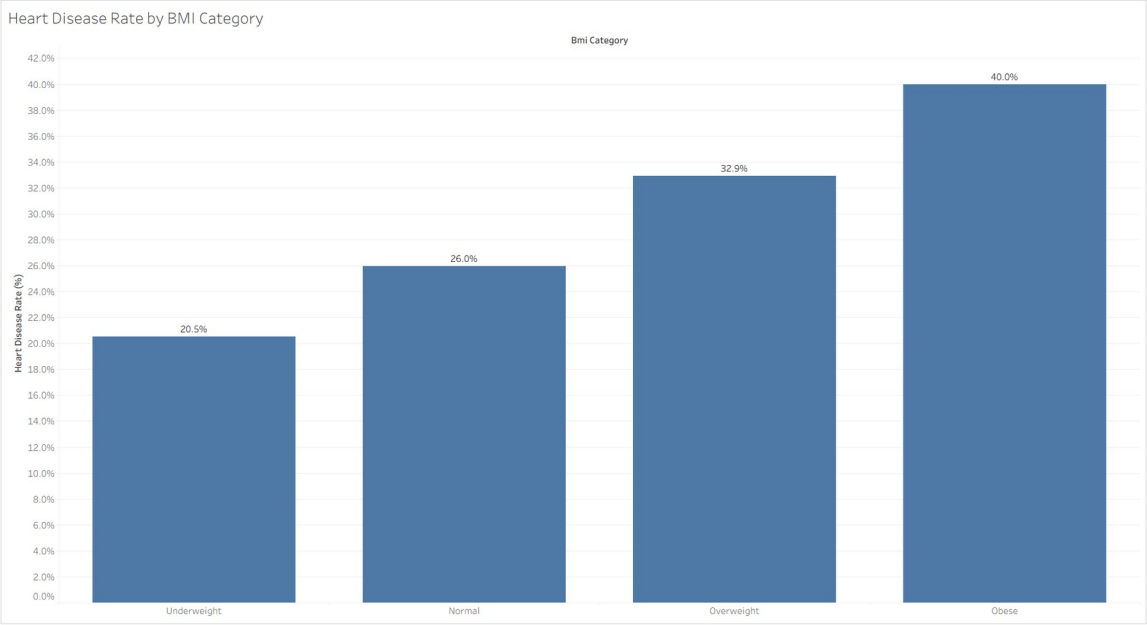
Insight: Counts are highest in the Normal (522 heart disease, 660 diabetes) and Overweight (550 and 580) categories because those categories have the most patients.



7. How does the heart disease rate change with BMI category?

Visualization: Bar chart of heart disease rate by BMI category.

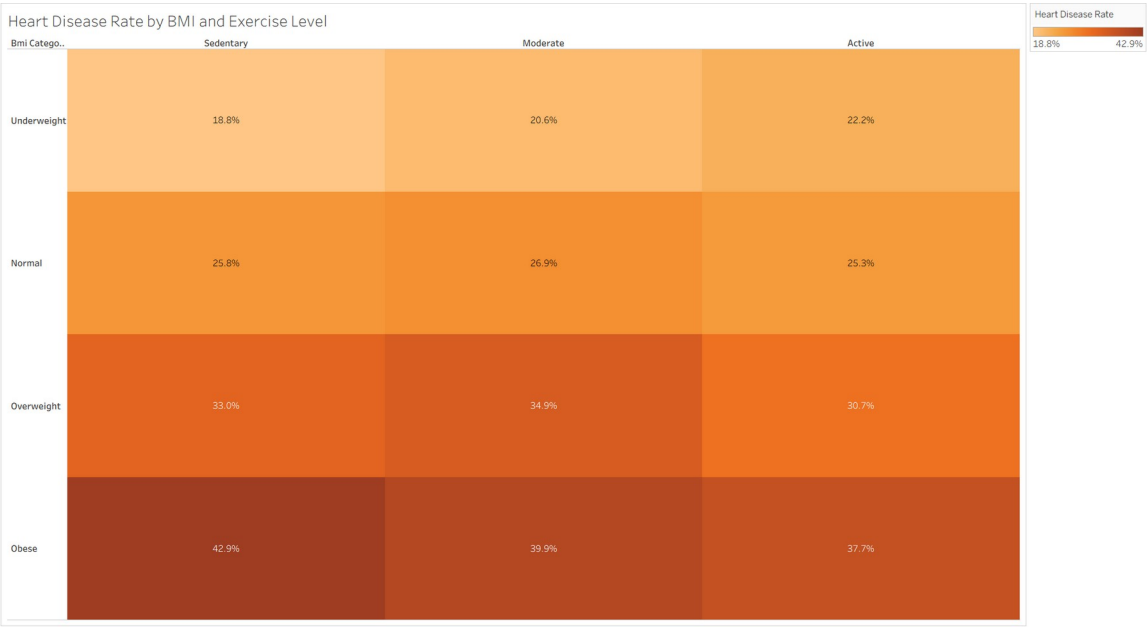
Insight: The rate rises steadily: Underweight 20.5%, Normal 26.0%, Overweight 32.9% and Obese 40.0%.



8. Which combination of BMI and exercise level has the highest heart disease rate?

Visualization: Heat map of heart disease rate by BMI category and exercise level.

Insight: Obese and Sedentary patients have the highest rate (42.9%); Underweight and Sedentary patients have the lowest (18.8%).



9. How does fatigue vary across age groups?

Visualization: Stacked bar chart of patients by age group and fatigue.

Insight: Fatigue is reported by 397 to 521 patients in every age group, and the Teenager group has the most (521).

